LEF MODEL · CHAPTER 26

Chapter 26: The Ten Further Requirements (R9–R18), the Hafnium–Tungsten Clock and the Map of Future LEF Research

Based on Paper 13 — 13. The Requirements (Parts C–D and Appendix B: requirements R9–R18, the flow of information between the papers, the Hf–W clock and the 7 research tasks)

01 · THE MYSTERY

The mystery: which signatures, unique to the Earth's pear alone, did the process leave on the Earth and on the Moon?

In Chapter 25 we checked the first eight requirements (R1–R8). But while features such as isotopic identity or a small core can receive alternative explanations in the literature, **Parts C and D of Paper 13** set out ten further requirements (**R9–R18**) that follow only from the geometry of the Earth's pear, its collapse back into the mantle and the exit conduits:

1. **The traces of the pear on the Earth and on the Moon (R9–R15, §C.1–C.6):** where did the 7–12 lunar masses of the pear's body sink after the detachment? How did the concentration of KREEP elements on the Moon's near side form? And why does the density of craters at the poles rise precisely as the crater is larger and older?
2. **The follow-on requirements after the pear (R16–R18, §C.7):** how do we prove that the Moon's 34 maria are ejection openings in which ancient

craters were melted, why is the centre of the South Pole–Aitken basin (SPA) shifted south relative to the two opposite openings, and how does Mars serve as an independent control test?

3. **The time constraint of the hafnium–tungsten clock (Hf–W, §D.2) and the map of future research:** how does a gradual build-up over 600 million years fit with a radioactive-decay clock with a half-life of only 8.9 million years?

02 · AN IMAGE

An image: a hard crust on slippery oil, the folded edges of a peel, and an hourglass that ran out early

Four tangible images illuminate the ten further requirements and the chronological constraint of the model:

- **A stiff shell sliding on a soft lubricating layer (pure compaction of craters, §C.1):** to preserve the ancient round craters carried on the crustal dome as the pear rose, the outer crust must stay very stiff ($\eta_{\mathrm{shell}} \gtrsim 10^{23}\ \mathrm{Pa \cdot s}$), while its base sits on a soft thermo-mechanical interface (the boundary between the light sial and the dense sima) that lets it slide and shrink in uniform hydrostatic compaction (Pure Compaction) without distorting the craters into ellipses!
- **The edges of a peel folded inward during peeling (the KREEP complex, R12):** when the Moon peeled off the base of the Earth's pear, the torn edges of the Earth's sial crust folded mechanically inward onto the circumference of the base on the near side (the Procellarum PKT region) and carried with them the incompatible elements (potassium K, rare earths REE and phosphorus P) — so the oldest anorthosites in the Moon's highlands must be depleted in potassium and thorium!
- **A heavy heap of iron sinking to the bottom of the pot (the LLVP lower anomaly in the Pacific, R10–R11):** the moment the Moon was released at the corotation line (16,399 km), the remaining body of the pear (~ 9 lunar masses, about 11% of the Earth's mass, all of it lying inside the corotation line, so no remnant of it was left in orbit) collapsed back into the mantle and sank to the core–mantle boundary directly beneath the base near Kiritimati island (1.9°N,

157.5°W) — creating the iron-rich lower anomaly of the Pacific (Pacific LLVP), while the African anomaly is the opposite lobe that was not disturbed!

- **An hourglass that emptied within tens of millions of years (the $^{182}\mathrm{Hf}$–$^{182}\mathrm{W}$ clock, §D.2):** the radioactive isotope $^{182}\mathrm{Hf}$ decays to $^{182}\mathrm{W}$ with a half-life of 8.9 million years. After 600 million years only a negligible remnant of 5.1 × 10^{-21} is left — that is, the hourglass stopped completely already at the start of the build-up window, so the tungsten signatures measured on the Moon today were locked at the beginning and carried over intact without being erased!

03 · SCIENCE AND MODEL

Science and model: the seven requirements of the pear (R9–R15) and the three follow-on requirements (R16–R18)

Parts C.6 and C.7 of Paper 13 detail all ten further requirements and separate those that have already received quantitative support in Papers 6–8 from those awaiting a final calculation:

1. **R9 and R10 — the equilibrium shape of the pear and its complete collapse:** the body of the pear (9,000 km wide at the base, converging to a contact of 3,000 km) is held as a hydrostatic-dynamic equilibrium shape (self-gravity against internal pressure and centrifugal force, not the tensile strength of rock) entirely inside the corotation line, so at detachment it collapses entirely inward without leaving any remnant in orbit.

2. **R11–R14 — the Pacific signature, the core gauge and the absence of foreign material:** the base of the pear at Kiritimati sits above the Pacific LLVP (R11); the Moon's tiny core (1.0%–2.0%) serves as a depth gauge showing how close the feeding stream came to the sinking iron (R13); and no foreign extraterrestrial component will be found in the bulk of the Moon or in the basalt plumes of the Pacific (R14).

3. **R15 (quantitatively supported in Paper 7) — compaction of craters at the poles depending on diameter:** the mechanical rounding compacted the

ancient craters toward the poles without reducing their linear diameter: the density ratio between the poles and the equator rises from **1.09** for craters 20–30 km across to **1.53** for ancient craters 60–100 km across!

4. **R16 and R17 (quantitatively supported in Papers 6 and 8) — maria as ejection openings and the SPA seal:** on the floor of the 34 active maria the craters that landed on liquid lava were melted, so the density of sharp craters (≥ 20 km) in them is **only 0.8 per million km²** against 18.3 on the plains and 207 in the highlands (R16); and in the South Pole–Aitken basin (SPA), the centre of the dome, the centre of heat and the crater minimum are shifted south (from 17°–21° in the flattened stage to 24°–29° today) relative to the two points opposite Imbrium and Serenitatis (R17).

5. **R18 (an independent control test in Paper 10) — the moons of Mars from Pavonis Mons:** the launch of Phobos and Deimos as corks from a single ejection conduit at Pavonis Mons (1.48°N) into a condensed steam ring; verification of the pure Martian composition awaits the samples of the MMX mission.

`The principle of isolating the levels (Multi-Level Constraint, §C.3 & §D.3)`

Paper 13 stresses that a mismatch in one morphological clause (such as the sorting of the ejection openings in §C.3 or the Mars control test in R18) is limited to that branch alone and does not rule out the corotation-line equation of Paper 2, which rests on the laws of conservation of mass and angular momentum.

04 · THE MECHANISM IN WORDS

The mechanism in words: research transparency, the constraint of the hafnium–tungsten clock and the seven open tasks

Part D, Appendix B and the future-research section of Paper 13 present a full scientific road map, separating with intellectual honesty existing knowledge from

the model's contribution and marking seven clear research targets:

1. **The chain of scientific control and ruling out a single point of success (§D.3):** to establish that a requirement *is not met*, a mathematical proof is required that there is no physical solution across the whole range of parameters and measurement errors (and not a single-point check). At the same time, one point of success does not confirm a whole path — every claim is subject to the chain: **prediction → mathematical detail → robustness to noise → accessibility for testing (the code package on Zenodo) → empirical test**.

2. **An explicit separation between existing physics and the novelty of the LEF model (Appendix B):** the paper states explicitly that the laws of gravity, mass conservation, differentiation, and the theory of ancient "Earthshine" (Roy et al. 2014) or the identification of the Procellarum rift rectangle (Andrews-Hanna et al. 2014) belong to their original owners. The unique novelty of LEF is **the double mechanical integration in the same body** — inner filling from below by a deep mantle stream, together with the lifting of the Earth's lithospheric shell from above, under a closed mass balance and centre-of-mass offset.

3. **The seven open tasks for future research (Further Work):** (1) completing the statistics of the concentric belts (T3, T4, and T1 on models 2–4 and GRGM1200B); (2) building a unified multi-system isotopic balance (O, Ti, Cr, W); (3) a multiphase flow model for transporting iron droplets to the core (1.4%); (4) a joint vector inversion of crustal thickness and the centre-of-mass offset in the true contact geometry; (5) a controlled chronological comparison of clock 1 in primary highland crust from both sides; (6) a quantitative classification of the 34 maria as ejection openings; and (7) an independent tectonic reconstruction of the Pacific lithosphere above the Pacific LLVP.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

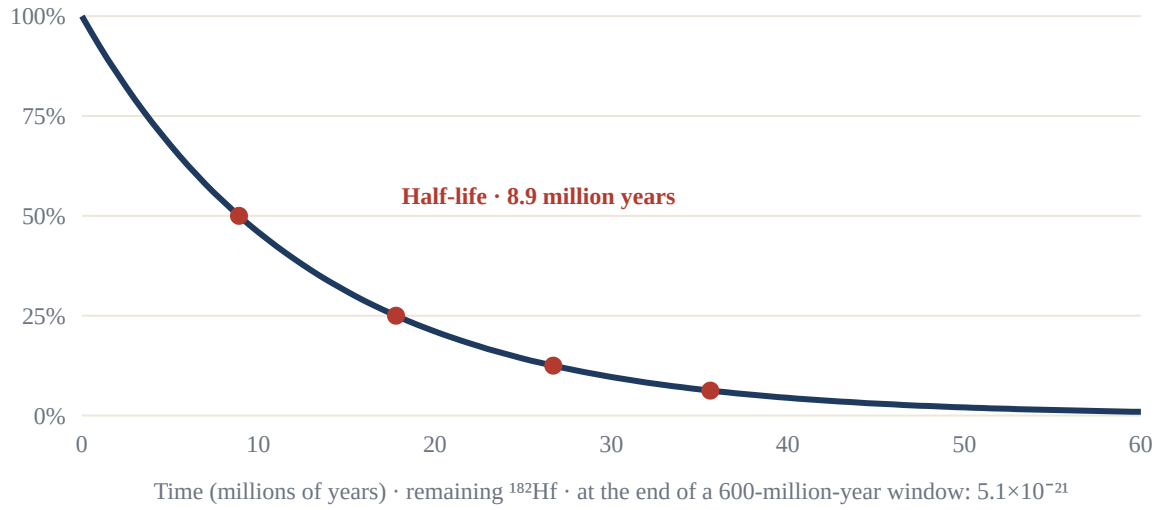
The radioactive-decay equation of the short hafnium–tungsten differentiation clock ($^{182}\mathrm{Hf} \rightarrow {}^{182}\mathrm{W}$, half-life $T_{1/2} = 8.9\ \mathrm{Myr}$) over the build-up window of 600 million years, and the methodological chain of verification, are formulated as follows (§D.2–D.3):

$$\frac{N(t)}{N_0} = 2^{-t/(8.9\ \mathrm{Myr})}$$

$$\frac{N(600\ \mathrm{Ma})}{N_0} \approx 5.1 \times 10^{-21}$$

$$\text{prediction} \longrightarrow \text{detailed} \longrightarrow \text{robust} \longrightarrow \text{assessable} \longrightarrow \text{testable}$$

Since the ratio $N(600\ \mathrm{Ma})/N_0 \approx 5.1 \times 10^{-21}$ leaves zero live $^{182}\mathrm{Hf}$ atoms at the end of the build-up, the tungsten clock cannot be reset at the end of the window — so every chronological model must preserve the tungsten signature set at the start of differentiation over 600 million years.



The hafnium–tungsten clock: after about 50 million years almost no $^{182}\mathrm{Hf}$ is left, so the tungsten signature is locked at the start of the build-up window and kept until its end.

The signature in reality: the flow of data through the series of papers and the open protocol on Zenodo

Part D.1 and the code section of Paper 13 bind the whole series of papers into a transparent system of dependencies that any outside researcher can fully reproduce (§D.1, Data & Code):

- **The flow of constraints from Paper 2 to Paper 13 (§D.1):** Paper 2 fixes the corotation radius at detachment (16,399 km in the equatorial plane, 17,287 km at latitude 22.5° for a day of 5.77 hours); Paper 3 sets a minimum transition radius of ~ 1,000 km carrying 0.46 lunar masses and a sideways shear load $F_{\text{shear}} \approx 10^{22}\ \text{N}$ (ruling out a "narrow neck") and a shell viscosity $\eta_{\text{shell}} \gtrsim 10^{23}\ \text{Pa}\cdot\text{s}$; and Papers 4–10 add the orbit, the maria, the craters, the poles, the SPA basin and Mars.
- **The open reproduction package (Zenodo DOI: 10.5281/zenodo.22743615):** all the calculation scripts in Python 3 (NumPy and SciPy), the input files and the outputs of tests T1, T2 and T5 are permanently deposited in the Zenodo repository for independent reproduction.

The lab box: the ten further requirements (R9–R18) and the mismatch criteria of the LEF model (Paper 13, Parts C–D)

Requirement	The physical condition required in the model	Test measure / mismatch outcome	Current status
R9 (pear in equilibrium)	A body 9,000 km wide converging to 3,000 km is held by self-gravity inside the corotation line	Mismatch: no mathematical equilibrium shape in the range of density and rotation	Open for a full equilibrium calculation
R10 (collapse of the pear)	The remnant of the pear (7–12 lunar masses, ~ 11% of the Earth's mass) collapses entirely inward with no remnant in orbit	Mismatch: identification of a remnant in orbit, or absence of a mass anomaly beneath the source	Open (consistent with the folds of Paper 9)

R11 (Pacific LLVP)	The base of the pear at Kiritimati (1.9°N, 157.5°W) sits above the iron-rich Pacific LLVP	Mismatch: the base outside the LLVP, or the two lobes (Pacific and African) identical	Open for a tomographic test
R12 (KREEP complex)	The edges of the Earth's sial folded inward fix KREEP in Procellarum; the anorthosites are depleted in K and Th	Mismatch: no geometric link to the edges of the base, or a sial signature in the anorthosites	Open for geochemical mapping
R13–R14 (core gauge and zero foreign material)	The core (1.4%) measures closeness to the sinking iron; zero foreign impactor component in the Moon and in the Pacific plumes	Mismatch: identification of an extraterrestrial chemical/isotopic component in the bulk of the Moon or in the plumes	Open for a multi-system balance
R15 (compaction of craters at the poles)	**The rounding compacted craters toward the poles in a ratio rising with diameter, without reducing their linear diameter**	**Pole/equator density ratio: 1.09 (20–30 km) → 1.53 (60–100 km)**	**Passed successfully (Paper 7)**
R16 (the 34 maria as ejection openings)	**The floor of the active maria melted ancient craters that landed on it**	**Density of sharp craters ≥ 20 km: 0.8 in the maria against 18.3 on the plains and 207 in the highlands**	**Quantitatively supported (Paper 6)**
R17 (two openings and a dome at SPA)	**The centre of the dome, the heat and the crater minimum at SPA are shifted south (24°–29°) relative to the points opposite Imbrium and Serenitatis**	**Mismatch: no geometric link, or a position north of the opposite points**	**Quantitatively supported (Paper 8)**
R18 (the moons of Mars from Pavonis)	Phobos and Deimos were launched from a single ejection conduit at Pavonis Mons (1.48°N) into a steam ring	Mismatch: a non-Martian composition in the MMX samples, or a mass excess beneath Pavonis	Open (an external control test, Paper 10)

07 · NEXT

The door ahead

In Chapters 25–26 we completed the survey of **Paper 13 (The Requirements)** — the methodological compass of the LEF series, gathering all 18 requirements (**R1–**

R18), the results of the statistical tests (**T1–T5**), the constraint of the hafnium–tungsten clock and the 7 open tasks for future research.

Chapter 27

The Research Map of the LEF Model (Part A) — the Early Earth, the Formation of the Moon and the SPA Basin (Topics 1–79) →

← Chapter 25 · Prologue

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